

A conjecture about direct product of funcoids

OPG-37540

L1 - TRACTABLE

PARTIALLY SOLVED

Topology

ORIGINAL PROBLEM STATEMENT

Conjecture Let f_1 and f_2 are monovalued, entirely defined funcoids with $\text{Src } f_1 = \text{Src } f_2 = A$. Then there exists a pointfree funcoid $f_1 \times^{(D)} f_2$ such that (for every filter x on A)

$$\langle f_1 \times^{(D)} f_2 \rangle x = \bigcup \{ \langle f_1 \rangle X \times^{\text{FCD}} \langle f_2 \rangle X \mid X \in \text{atoms}^{\mathfrak{A}} x \}.$$

(The join operation is taken on the lattice of filters with reversed order.)

A positive solution of this problem may open a way to prove that some funcoids-related categories are cartesian closed.

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